MILITARY SPECIFICATION

FUZE, ROCKET: M433
LESS BOOSTER AND BOOSTER LEAD

This Amendment forms a part of Military Specification MIL-F-63281 (AR), dated 29 September 1978, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, STANDARDS: Add:

"MIL-STD-1167 - Ammunition Data Card

PAGE 2

2.1, PRODUCT AND PACKAGE DRAWINGS: Delete: 8886458, 9260924 and 9260925 in their entirety, and substitute:

"8861213	Box Wirebound, Packing Ammunition for Artillery Type or Rocket Fuzes
9230113	Container Ammo, Fiber PA30 for Rocket 2.75 Inch: (Warheads M151, M156, XM157, XM158, XM282 and M230)
9230114	Box Packing, Ammunition for Rocket 2.75 Inch: (Warheads M151, M156, XM157, XM158, M230, XM232 and M247)
8882186-3	Warhead, 2.75 Inch Rocket, HE M151/without Fuze"

FSC 1340

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PAGE 3

3.3.4: Delete in its entirety.

3.3.5: Delete in its entirety and substitute:

"3.3.5 Top plate assembly. Dwg. 9332603.

"3.3.5.1 Conductor strip rivets. The rivets shall withstand a minimum straight push-out force of one (1) pound applied to the swaged end."

PAGE 4

3.4.1: Delete in its entirety and substitute:

"3.4.1 Module time delay function - (drawings 9260685, 9260878).

"3.4.1.1 Fixed voltage test.

a. After shorting pins 5, 6 and 4 together for four seconds, the electronic module shall be capable of accepting a setting charge and exhibiting a corresponding time delay as listed in Table IA. The positive voltage applied to Pin 1 referenced to Pin 4 shall be followed by the negative voltage as shown in Table IA, Vc1 and Vc4, respectively. The time of applied voltage shall be 6 msec $\pm$ 1% for Vc1 and 40 msec + 1% for Vc4. These voltages will be supplied thru a 47 ohm series resister to Pin 1 with reference to ground connected to Pin 4. The time delay shall be the time interval from the time Pin 2 is connected to Pin 4 and the leading edge of the output pulse exits Pin 3. The firing output (Pin 3) pulse shall provide at least 1000 ergs of energy in less than 100 microseconds into an M84 detonator or a resistor of 3.5 $\pm$ 1.5 ohms. Record the input voltages, time interval and energy output (greater than, equal or less than) by serial number of module. The 20 msec, 200 msec and 600 msec settings shall be used for production information tests on lot 1 through lot 4. All other settings are for lot 1 information only.

b. A seven (7) second time delay shall follow the charging of each module prior to initiating the module timing circuit.

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TABLE IA

Time Delay Setting (msec)	Capacitor Inputs (Volts) Vc1*(Timing)	Vc4*(Firing)	Function Time Delay Output (msec)
20	0.93	-15.07	17 to 23
50	1.57	-14.43	43 to 57
100	2.58	-13.42	87 to 114
200	4.33	-11.67	174 to 228
300	5.83	-10.17	260 to 343
400	7.08	- 8.92	346 to 458
500	8.11	- 7.89	432 to 575
600	8.98	- 7.02	517 to 695

*Initial values of Vc1 and Vc4 shall not vary by more than ± 1% from the nominal values shown.

"3.4.1.2 Compensated voltage testing.

a. A compensated voltage setter that represents the helicopter setter systems shall be used for delay time acceptance tests for both first article and lot acceptance tests.

b. After shorting pins 5, 6 and 4 together for four seconds the electronic module shall be capable of accepting a setting charge from the compensated voltage setter and exhibiting a corresponding time delay as listed in Table IB. The time delay shall be the time interval from the time pin 2 is connected to pin 4 and the leading edge of the output pulse exits pin 3. The firing output (Pin 3) shall provide at least 1000 ergs of energy in less than 100 microseconds into an M84 detonator or a resistor of 3.5 ± 1.5 ohms. Record the time delay setting, function time delay and energy output (equal to, greater than or less than) by serial number or module lot no. The 20 msec, 200 msec and 600 msec settings shall be used for all first article and production lot acceptance tests.

TABLE IB

Time-Delay Setting (msec)	Function Time Delay Output (msec)
20	5 - 35
200	185 - 215
600	570 - 630"

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PAGE 6

TABLE II: Add to table under indicated headings:

Position of S&A Dev.	Terminals	Requirements
Armed	#4 & #6 & rotor	100 milliohms Max.

3.6.4 At the end of the paragraph add the following sentence: "This test may be performed with a simulated detonator (unloaded S&A)."

PAGE 7

3.8.2: Delete in its entirety and substitute:

"3.8.2 Charge circuit continuity. (Fuze Unarmed) - The continuity of the charge circuit, through its umbilical, shall be 220 ohms minimum and 3.3k ohms maximum."

3.8.4.1a: Delete "-60° ± 5°F." and substitute "-60° ± 5°F. (-40° ± 5°F for Ballistic Firing Test)"

3.8.4.1b: Delete in its entirety.

PAGE 8

Add new paragraph 3.8.5.5:

"3.8.5.5 Five-foot drop. The fuze shall be safe to handle, fire or dispose of following the five-foot drop test. It need not be operable."

3.9: Delete after tests: "Five-Foot Drop Test".

* Delete "Vibration Test, (See Appendix A)" and substitute "Vibration Test, (See Appendix A); Transportation Vibration Test, Procedure 1, Test 119, of MIL-STD-331."

PAGE 9

3.13: Delete after inspection "(,See Table III)".

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4.3.1, Submission, List of Drawings:

Line 10, Rotor Assembly: Delete "8883744" and substitute "9260769".

Line 14 & 21: Delete "Contact Female" and "Contact Female #3" in their entirety.

Line 23, Case Assembly: Delete "9260756" and substitute "9332604".

Line 24, Case: Delete "9260758" and substitute "9332602".

Line 25, Top Plate Assembly: Delete "9260760" and substitute "9332603".

Line 28, Molded Electronics Assembly: Delete in its entirety.

* Line 35 & 36, Fuze Rocket, M433, Less Booster and Booster Lead: Delete "228" and substitute "366".

Line 39, Electronic Module Assembly: Delete "702" and substitute "25".

Add the following to table under indicated headings:

"PART DESCRIPTION	DRAWING	QUANTITY
Molded Electronics Assembly (Module)	9260878	702
Environmental Protector	9329847	25
Rivet	9331168	25
Contact	9332596	25
Contact	9332597	25
Contact	9332598	25
Contact	9332599	25
Contact	9332600	25"

4.3.2: Delete "IV" and substitute "VI".

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PAGE 12

Table III:

Sheets: Delete "1 of 8" and substitute "1 of 9".

Examination or Test:

Device, Safety and Arming w/o Case Assembly: Delete "Load Test of Detonators" in its entirety.

M433 S&A Assembly (Loaded):

Add: "(a)" after "Non-Arming".

Add: "(a)" after "S&A Arming Time".

Add: "Load Test of Detonator, 25, 3.74, 4.5.9".

Notes: Add "(a) This test may be performed with a simulated detonator (unloaded S&A)."

PAGE 13

Table III:

Sheets: Delete "2 of 8" and substitute "2 of 9".

PAGE 14

Table III:

Sheets: Delete "3 of 8" and substitute "3 of 9".

Examination or Test:

Lever Assembly: Delete "8883706" and substitute "9260776".

Rotor Assembly: Delete "8883744" and substitute "9260769".

Inner Plate Assembly: Delete "8883693" and substitute "9260804".

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PAGE 15

Table III:

Sheets: Delete "4 of 8" and substitute "4 of 9".

Examination or Test:

Contact Female: Delete in its entirety.

Contact Female #3: Delete in its entirety.

PAGE 16

Table III:

Sheets: Delete "5 of 8" and substitute "5 of 9".

Examination or Tests:

Delete: "Molded Electronics Assembly (Dwg. 9260878)" and substitute "Electronic Module Assembly (Dwg. 9260887)".

Molded Electronics Assembly (Dwg. 9260878): In paragraph reference delete "4.4.2.21" and substitute "4.4.2.28".

Case Assembly: Delete "9260756" and substitute "9332604". In paragraph reference delete "4.2.2.23" and substitute "4.4.2.23".

PAGE 17

Table III:

Sheets: Delete "6 of 8" and substitute "6 of 9".

Examination or Test:

Top Plate Assembly: Delete "9260760" and substitute "9332603".

Delete: "Pin Retention of 0.25 in Dia. Pin Test" and substitute "Rivet Push Out Test".

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Delete: "Electronics Assembly (Module) (Dwg. 9260887)" and substitute "Molded Electronics Assembly (Dwg. 9260878)".

Paragraph Reference: Delete "4.4.2.28" and substitute "4.4.2.21".

PAGE 18

Table III:

Sheets: Delete "7 of 8" and substitute "7 of 9".

Examination or Test:

Fuze, Rocket, M433 Less Booster and Booster Lead:

* Examination for Defects: Delete "228" and substitute "366".

* S&A Position Check: Delete "228" and substitute "366".

Five Foot Drop - MIL-STD-331, Test III: Delete "3.8" and substitute "3.8.5.5"

* Add "Transportation Vibration 120 3.9 4.5.38".

Aircraft Vibration - Helicopter: Delete "63" and substitute "81".

PAGE 19

Table III:

Sheets: Delete "8 of 8" and substitute "8 of 9".

Examination or Test:

Delete: "Leak Test" in its entirety.

Ballistic Firing Test: Delete "144 and substitute "138".

* Charge Circuit Continuity: Delete "228" and substitute "366".

Add the following below "Charge Circuit Continuity":

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"Mechanical Interrupt 32 3.8.5.2 4.5.23/4.5.38

Case: Delete "9260758" and substitute "9332602".

Add included page 24.

PAGE 21

4.4.2.1:

Major Defect 101: Delete "Forward Face" and substitute "Upper Surface".

Major Defect 107: Delete in its entirety.

Major Defect 110: Delete "across posts" and substitute "across outside corners of posts".

Major Defect 113: Delete in its entirety.

Major Defect 114: Delete in its entirety.

Major Defect 115: Delete in its entirety.

Notes: Delete "(a)" in its entirety.

PAGE 22

4.4.2.1, Major 116: Delete "across the rotor housing posts".

PAGE 23

4.4.2.2:

Critical Defect 1: Add "(e)".

Major Defect 102: Delete in its entirety and substitute the following:

"102. Less than two speed clip tabs retaining detonator or no speed clip tabs in at least point contact with end of detonator, or detonator protrudes beyond rotor outside surface".

Major Defect 104: Delete "Gage" and substitute "Gage/Dwg. Note 7".

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Add Major Defects 107, 108, 109, 110, 111 and 112 as follows:

Category	Examination or Test	AQL Or 100%	Requirement Paragraph	Paragraph Reference
107	Detonator perforated or damaged	0.40%	3.3	Visual
108	Load test of detonator	0.40%	3.7.4	4.5.9
109	Arming time (e)	100%	3.7.2	4.5.10(d)
110	Distance from detonator disc to bottom of S&A Rotor Housing (armed position)	0.40%	3.3	Gage
111	Outside diameter	0.40%	3.3	Gage
112	Contact resistance between #4 and #6 contacts and rotor (armed, less detonator)	100%	3.6.3	Milliohm-meter & Gage"

Add the following NOTES:

"(c) Detonators are to be tested in accordance with 4.5.2.
(d) To be performed at ambient temperature only.
(e) May be performed with a simulated detonator."

PAGE 30

4.4.2.9: Delete Drawing Number "8883706" and substitute "9260776".

PAGE 31

4.4.2.10:

Delete: Drawing Number "8883744" and substitute "9260769".

Major Category: Delete "None Defined".

Add New Major Defects 101 and 102 as follows:

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"101. Banking Stud Pullout Test 0.40% 3.3 Gage

102. Rotor Ground Strap and Detonator Socket Contact Resistance 0.40% 3.3 Milliohmmeter."

PAGE 32

4.4.2.11:

Delete: Drawing Number "8883693" and substitute "9260804".

Major Cetegory: Delete "None Defined".

Add New Major Defect 101 as follows:

"101. Pillar Push Test 0.40% 3.3 Gage."

PAGE 33

4.4.2.12: Delete Drawing Number "8883692" and substitute "9260774".

PAGE 35

4.4.2.14: Delete in its entirety and substitute included Page 25.

PAGE 43

4.4.2.21:

Major Category: Add New Major Defects 108, 109, 110, 111, 112 and 113 as follows:

"108. Function Time Delay: 20 Msec 100% 3.4.1.2 4.5.35.

109. Function Time Delay: 200 Msec 100% 3.4.1.2 4.5.35.

110. Function Time Delay: 600 Msec 100% 3.4.1.2 4.5.35.

111. Module Turn on Time 0.40% 3.4.2 4.5.31

112. Thermal Shock 30 3.8.4 4.5.30b

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113. Maximum Envelope Distortion 0.40% 3.3 Gage".

Minor Category: Delete Minor Defects 204 and 205 in their entirety.

PAGE 44

* 4.4.2.22, add Critical 1: "1 Switch tip profile (taper and radius) 100% 3.3 Gage".

* Add Major 107: "107 Resistance between switch tip and weight .40% 3.3 Megohmeters @ 500 VDC".

PAGE 45

4.4.2.23: Delete in its entirety and substitute included Page 26.

PAGE 46

4.4.2.24: Delete DRAWING NUMBER "9260758" and substitute "9332602".

Delete NEXT HIGHER ASSEMBLY "9260756" and substitute "9332604".

Major Defects 101 and 105: Delete in their entirety.

PAGE 47

4.4.2.25: Delete in its entirety and substitute included Page 27.

PAGE 49

4.4.2.27: Delete Major 101 in its entirety.

PAGE 50

4.4.2.28: Delete in its entirety and substitute included Page 28.

PAGE 52

4.4.2.30, Major Category:

Major Defect 108: Delete in its entirety and substitute "True position of small diameter to large diameter of the umbilical hole."

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Major Defect 109: Delete in its entirety and substitute "True position of intermediate diameter to large diameter of the umbilical hole."

PAGE 53

4.4.2.30, Major 117: Delete "stem" and substitute "stem cavity".

PAGE 55

4.4.2.32, Major 103: Delete "Uniform wall thickness" and substitute "Wall thickness not uniform".

PAGE 56

4.4.2.33:

Critical Defect 4: Delete "15" and substitute "6".

Critical Defect 5: Delete "54" and substitute "63".

Add New Critical Defect 10 and 11 as follows:

"10 Mechanical Interrupt 32(a) 3.8.5.2 4.5.23/4.5.38

* 11 Transportation vibration 30 3.9 4.5.39"

Major Defect 103: Delete in its entirety.

* Major defect 105: Delete "60" and substitute "90".

Major Defect 106: Add "(c)" after "Gage".

Add New Major Defect 107 as follows:

"107. Charge Circuit Continuity (following helicopter vibration) 54 3.8.2 Gage (c)"

Add New Major Defect 108 as follows:

* "108. Charge Circuit Continuity (following transportation vibration) 120 3.9 Gage (c)".

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Add Note (c) as follows:

"The instrument to be used shall meet the following criteria: (1) Output pulsewidth 100 ± 10 msec; (2) Output impedance 150K ohms typical; (3) Output open circuit voltage 100 ± 10 millivolts".

PAGE 57

4.4.2.34:

Major Defect 104: Delete in its entirety.

Minor Defect 203: Delete in its entirety.

Note: Delete in its entirety.

PAGE 59

4.4.2.36: Delete in its entirety and substitute included Page 29.

PAGE 60

* Delete in its entirety and add included Pages 30 to 34.

PAGE 64

4.5.6.1: Delete in its entirety and substitute:

"4.5.6.1 Load test of No. 1 gear and pinion assembly. Apply the axial load perpendicular to the No. 1 gear and in such a direction as to test the assembly of the No. 1 gear and pinion. The required force shall act against the No. 1 gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any No. 1 gear and pinion assembly that fails the requirement shall be classed defective."

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4.5.6.3: Delete in its entirety and substitute:

"4.5.6.3 Load test of escapement gear and pinion assembly. Apply an axial load perpendicular to the escapement gear and in such a direction as to test the assembly of the escapement gear and pinion. The required force shall act against the escapement gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any escapement gear and pinion assembly that fails the requirement shall be classed defective."

PAGE 65

4.5.7: Delete in its entirety.

PAGE 66

4.5.10: Delete in its entirety and substitute:

"4.5.10 M433 S&A arming tests ambient and hot. The S&A in the safe position shall be assembled to the test equipment in accordance with 4.4.4. The g-weight shall be to the right when looking from the outside of the centrifuge spin table toward the center of rotation of the spin table. The equipment shall accelerate the Safety and Arming Device to 40.0g's $\pm$ 0.5g's in a direction which simulates rocket launch. The arming time shall be recorded to the nearest hundredth of a second using a calibrated timer. At the completion of acceleration, the equipment shall be halted and visual observation shall be made for the locking position of the detent to ascertain positive arming. At the completion of the test, the Safety and Arming Device shall be returned to the safe position. Any S&A that fails the requirement shall be classed defective and removed from the lot.

"WARNING: This test may be dangerous, and should be performed behind a suitable barricade."

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PAGE 67

4.5.13: Delete in its entirety and substitute:

"4.5.13 M433 S&A arming time test - cold. The S&A in the safe position shall be assembled to the test equipment in accordance with 4.4.4. The setback weight shall be to the right when looking from the outside of the centrifuge spin table toward the center of rotation of the spin table. The equipment shall accelerate the loaded Safety and Arming Device in a clockwise direction to 27.0g's plus or minus 0.5g's and the arming time shall be recorded to the nearest hundredth of a second using a calibrated timer. S&A's failing to comply with the specified requirements shall be classed defective."

4.4.14: Delete in its entirety and substitute:

"4.5.14 M433 S&A non-arming test. The S&A in the safe position shall be assembled to the test equipment in accordance with 4.4.4. The setback weight shall be to the right when looking from the outside of the centrifuge spin table toward the center of rotation of the spin table. The equipment shall accelerate the loaded Safety and Arming Device in a clockwise direction to 12.0g's minimum for a minimum of five (5) seconds. Observation shall be made for rotor in the safe position. S&A's failing to comply with the specified requirement shall be classed defective and removed from the lot."

PAGE 68

4.5.16: Delete in its entirety and substitute:

"4.5.16 M433 fuze assembly arming time test. The fuze assembly shall be assembled to the equipment in accordance with 4.4.4. The setback weight shall be to the right when looking from the outside of the centrifuge spin table toward the center of rotation of the spin table. The equipment shall accelerate the loaded fuze in a clockwise direction to 40.0g's plus or minus 0.5g's except for cold temperature which shall be done at 27g's plus or minus 0.5g's and the arming time shall be recorded to the nearest hundredth of a second using a swing fixture and acoustic pick-up feeding the arming cycle noise signal to a design agency approved recorder. Any fuze assembly that fails the requirements of the arming time test shall be classed defective."

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4.5.17: Delete in its entirety and substitute:

"4.5.17 M433 fuze assembly non-arming test. The fuze assembly shall be assembled to the equipment in accordance with 4.4.4. The g-weight shall be to the right when looking from the outside of the centrifuge spin table toward the center of rotation of the spin table. The equipment shall accelerate the loaded fuze assembly in a clockwise direction to 12.0g's plus or minus 0.5g's for a minimum of three (3) seconds. An examination shall be made for rotor in the safe position. Any fuze assembly that fails the requirements of the non-arming time test shall be classed defective."

4.5.18:

Sentence one: Delete "The fuze assembly with warhead and motor" and substitute "The fuze assembly with inert booster, inert booster lead, XM230 warhead and inert motor".

Sentence two: Delete "8886810 and 8886811" and substitute "9230113 and 9230114".

PAGE 69

4.5.20, line 2: Delete "fifteen (15)" and substitute "six (6)".

4.5.20a:

Line one: Delete "The fuze assembly will be assembled to an M230 Warhead" and substitute "The fuze assembly will be assembled to an inert booster, inert booster lead and M230 Warhead".

Second paragraph: Delete "Five (5)" and substitute "Two (2)".

4.5.20c: Delete in its entirety and substitute:

"c. The fuzes shall be subjected to the tests and acceptance or rejection criteria as specified in Table IV. Fuzes must be safe to handle and safe to fire or dispose of (not necessarily operable) following the Five Foot Drop Test and prior to performing the Non-Arm and Arming Time Tests.

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TABLE IV

Test	Accept	Reject	Test Method Paragraph
Non-Arm	0	1	4.5.17
Arming Time			4.5.16
Safety Failure	0	1	4.5.24
Under Min.	0	1	4.5.24

Any fuze that is unsafe to handle, fire or dispose of following the Five Foot Drop Test or does not meet the arming time safety definitions of 4.5.24 is classified as a safety failure. If the sample fails the above acceptance criteria, the lot shall be rejected."

PAGE 70

4.5.20d: Delete in its entirety.

4.5.22: Delete after vibration "the fuzes shall be subjected to the seal Leakage Test, 4.5.7 and the lot shall be rejected if four (4) or more defects occur. Following seal leakage testing,".

Delete the Note in its entirety and substitute:

"Note: The fuzes shall be subjected to and must pass the continuity test following the vibration test. If one or more continuity failures occur in the test sample the lot shall be rejected."

4.5.23: Add to TABLE V under indicated headings:

Test	Accept	Reject	Test Method Paragraph
Mechanical interrupt	0	1	4.5.38

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PAGE 71

4.5.24.4: Delete in its entirety and substitute:

"4.5.24.4 Defect identification. Identification of defects as non-arm failures, arming time safety failures, arming time under minimum failures will depend on both centrifuge testing and umbilical continuity examination. In addition, items with these failures shall be removed and not subjected to the firing test."

PAGE 73

4.5.25, last paragraph: Delete "QAR-M" and "QAR-I" and substitute "QAF-T" and "QAF-I", respectively.

PAGE 74

4.5.26, NOTE 1: Delete "QAR-I" and "QAR-M" and substitute "QAF-I" and "QAF-T", respectively.

4.5.27: Delete in its entirety and substitute:

"Rivet push-out test. The resistive force shall act against the plate while avoiding support of the conductor strip. Apply the force with a 0.047 ± .003 diameter pin. Any rivet that pushes out shall be classed defective."

PAGE 75

4.5.30a: Delete in its entirety and substitute:

"a. The fuze shall be subjected to the Thermal Shock Test in accordance with MIL-STD-331, Test 113. These fuzes shall then be centrifuge fired. The fuze shall be charged to the nominal 600 msec time before arming. After arming, the sensor probe should be activated and the fuze fired. Any fuze that does not meet test requirements shall be classified defective."

4.5.31: Delete "3.4.1" and substitute "3.4.2".

4.5.36: Delete in its entirety and substitute:

"4.5.36 Ballistic firing test. The loaded M433 fuze shall be conditioned in accordance with Table VI or Table VII, whichever is applicable. This test shall be performed at a Government facility using M230 warheads with photoflash

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cartridges for the bunker and canopy tests and M151 warheads for the superquick tests. MK40, Mod 3 rocket motors shall be used for all tests and they shall be temperature preconditioned to +155°F ± 4°F prior to assembling to warhead/fuze. The fuze shall be set with a setter that represents the helicopter setting system. A hard copy of the fuze charge signature shall be made at this time. Prior to setting and firing, the fuzes shall be checked for charge circuit continuity. Fuzes that fail the continuity tests shall not be subjected to the ballistic test. Range markers shall be situated along the flight path. Airburst range behind the target shall be detected and recorded.

"4.5.36.1 Superquick test. The SQ setting shall be used for this test. The launcher shall be so located that rockets will impact at a slant range of approximately 3000 feet when launched at a QE of 0 to -20°. A QE of -7° will be used unless otherwise specified by the procuring agency. The impact area shall be such that a rocket launched from this altitude shall not have a trajectory resulting in a slant range in excess of 4000 feet. Impact shall cause the warhead to detonated high order.

"4.5.36.2 Simulated bunker test. A 20 msec setting shall be used for this test. The rounds shall be tested against a 30-to 40-inch thick bunker made of nominal 1-foot square logs or equivalent and located at a distance of 1100 feet from the launcher. Velocity at target impact will be about 1700 feet per second.

"4.5.36.3 Simulated forest penetration (canopy) test. The 200 and 460 msec settings shall be used for this test. The rounds shall be tested against a 20-foot high by 40-foot wide wall of 3/4 inch thick plywood located at a distance of 1100 feet from the launcher. Velocity at target impact will be about 1700 feet per second.

"4.5.36.4 No-test criteria. Some ballistic firings may result in conditions that do not properly test fuze function in the bunker and canopy operating modes. If the following conditions occur during a test firing, that launch will be scored as a "no test":

a. The rocket misses the target.

b. The rocket hit the ground before reaching the target.

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c. The rocket impacts the target, but hits the ground before the maximum acceptable range behind the target is reached.

d. The rocket hits the target supports.

e. The rocket passes through a hole in the target made by a previous firing.

"4.5.36.5 Acceptance criteria for operating mode tests.

Nominal Time Setting	Target	Acceptance Criteria
Superquick	Natural Terrain	Function on impact.
20msec	Bunker	Surface contact to 85 feet behind the target.
200msec	Canopy	Function at 25 to 475 feet after first target contact.
460msec	Canopy	Function at 50 to 1075 feet after first target contact.

4.5.36.6 Rejection criteria. For scoring the ballistic test, umbilical failures and improper charging signals shall not be charged to any individual operating mode test of 4.5.36.5."

* 4.5.36.6.1, Delete in its entirety and substitute:

"4.5.36.6.1 First article sample. If five (5) or more units do not meet the requirements in any one of the operating mode tests of 4.5.36.5, or four (4) or more units fail the charge circuit continuity tests, or sixteen (16) or more units fail the overall ballistic test (Table VI), the first article ballistic sample shall be rejected."

* 4.5.36.6.2, Delete in its entirety and substitute:

4.5.36.6.2 Lot Acceptance sample. If three (3) or more units do not meet the requirements in any one of the operating mode tests of 4.5.36.5, or two (2) or more units

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fail the charge circuit continuity tests, or seven (7) or more units fail the overall ballistic test (Table VII), the lot shall be rejected."

PAGE 76

Add 4.5.38:

"4.5.38 Mechanical interrupt test. The armed fuze assemblies from 4.5.16 shall be checked for charge circuit continuity using the gage described in Note c of 4.4.2.33. The gage shall indicate an open circuit between the charge circuit and the umbilical. Any fuze assembly that fails the requirement shall be classed defective."

Add 4.5.39:

* "4.5.39 Transportation vibration. The fuze shall be subjected to the Transportation Vibration Test in accordance with MIL-STD-331, Test 119, Procedure 1. The fuze shall be subjected to and must pass the continuity test following the Transportation Vibration Test. If one or more continuity failures occur in the test sample, the lot shall be rejected."

Add 5.3:

"5.3 Ammunition data cards. Data card information shall be as specified in MIL-STD-1167."

PAGE 77

6.5.2: Delete in its entirety and substitute:

"6.5.2 Boosters, booster leads and warheads.

a. The procuring agency will provide inert boosters, inert booster leads and M230 warheads for the five-foot drop test.

b. The procuring agency will provide inert boosters, inert booster leads, M230 warheads and inert rocket motors for the forty-foot drop test.

c. The procuring agency will provide boosters, booster leads, M151 warheads, M230 warheads, photoflash charges and rocket motors for the ballistic firing test."

6.6: Delete "QAR-M" and substitute "QAF-T".

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PAGE 78

6.8: Data Requirements, Paragraph headings: Delete "4.5.39" and substitute "4.5.37".

PAGE 79

* TABLE VI: Delete in its entirety and substitute included Page 35.

PAGE 80

TABLE VI (Continued): Delete in its entirety and substitute included Pages 36 and 37.

PAGE 81

* TABLE VII: Delete in its entirety and substitute included Page 38.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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TABLE III. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 9 OF 9		DRAWING NUMBER
	Fuze, Rocket, M433, Less Booster and Booster Lead and Components				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Rivet (Dwg. 9331168) Examination for Defects	25		3.3	4.4.2.37
	Contact (Dwg. 9332596) Examination for Defects	25		3.3	4.4.2.14
	Contact (Dwg. 9332597) Examination for Defects	25		3.3	4.4.2.36
	Contact (Dwg. 9332598) Examination for Defects	25		3.3	4.4.2.38
	Contact (Dwg. 9332599) Examination for Defects	25		3.3	4.4.2.39
	Contact (Dwg. 9332600) Examination for Defects	25		3.3	4.4.2.40
	Environmental Protector (Dwg. 9329847) Examination for Defects	25		3.3	4.4.2.41
NOTES:					

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Contact				9332596
					NEXT HIGHER ASSEMBLY
					9332603
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Angle of bend (58° + 2°)		0.40%	3.3	Gage
102	Thickness of strip		0.40%	3.3	Gage
103	Poor quality of plating		0.40%	3.3	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.14	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.23	Case Assembly				9332604
					NEXT HIGHER ASSEMBLY
					9260750
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Symmetrical location of detonator contact		0.40%	3.3	Visual
102	Contact resistance between #1 & #5 contacts (safe) (a)		100%	3.6.2	Milliohmmeter & Gage
103	Insulation resistance of #1 & #2 contacts (safe) (a)		100%	3.6.3	Megohmmeter, test voltage 500 VDC
104	Continuity through detonator (a)		0.40%	3.7.1	Gage
105	Insulation resistance of #1 & #5 contacts (armed) (a)		100%	3.6.3	Megohmmeter, test voltage 500 VDC
Minor					
201	Evidence of poor workmanship		0.65%	3.14	Visual

NOTES:
(a) Majors 102 to 105 are predicated on using a gage simulating a safe/armed S&A, incorporating thest features:
1. A setback weight with a switch actuator pin.
2. A rotor with a detonator socket grounded by a strap.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9332603
4.4.2.25	Top Plate Assembly				NEXT HIGHER ASSEMBLY 9332604
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Conical end of any rivet protrudes from plate (5 places)		0.40%	3.3	Visual
102	Alignment of N.C. contact		0.40%	3.3	Visual
103	Alignment of common contact		0.40%	3.3	Visual
104	Alignment of N.O. contact		0.40%	3.3	Visual
105	Loose connectors		0.40%	3.3	Visual
106	Push-out test of 5 rivets		0.40%	3.3.5.1	4.5.27
Minor					
201	Evidence of poor workmanship		0.65%	3.14	Visual

NOTES:
N.C. = Normally closed
N.O. = Normally open

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.28	Electronics Module Assembly		9260887
			NEXT HIGHER ASSEMBLY 9260886

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of contact cup to diameter "A"		0.40%	3.3	Gage
102	Height of soldered pin from top (near surface) of molded module, min.		0.40%	3.3	Gage
103	Improper soldering		0.40%	3.3	Visual
Minor					
201	Improper bonding of contact		0.65%	3.3	Visual
202	Evidence of poor workmanship		0.65%	3.14	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9332597
4.4.2.36	Contact				NEXT HIGHER ASSEMBLY 9332603
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Angle of bends (58° + 2°, 98° + 2°)		0.40%	3.3	Gage
102	Thickness of strip		0.40%	3.3	Gage
103	Poor quality of plating		0.40%	3.3	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.14	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9331168
4.4.2.37	Rivet				NEXT HIGHER ASSEMBLY 9332603
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Poor quality of plating		0.40%	3.3	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.14	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9332598
4.4.2.38	Contact				NEXT HIGHER ASSEMBLY 9332603
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Angle of bend (30° + 2°)		0.40%	3.3	Gage
102	Thickness of strip		0.40%	3.3	Gage
103	Poor quality of plating		0.40%	3.3	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.14	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9332599
4.4.2.39	Contact				NEXT HIGHER ASSEMBLY 9332603
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Angle of bends (58° + 2°, 2 places)		0.40%	3.3	Gage
102	Thickness of strip		0.40%	3.3	Gage
103	Poor quality of plating		0.40%	3.3	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.14	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.40	Contact				9332600
					NEXT HIGHER ASSEMBLY
					9332603
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Angle of bend (58° + 2°)		0.40%	3.3	Gage
102	Thickness of strip		0.40%	3.3	Gage
103	Poor quality of plating		0.40%	3.3	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.14	Visual

NOTES:

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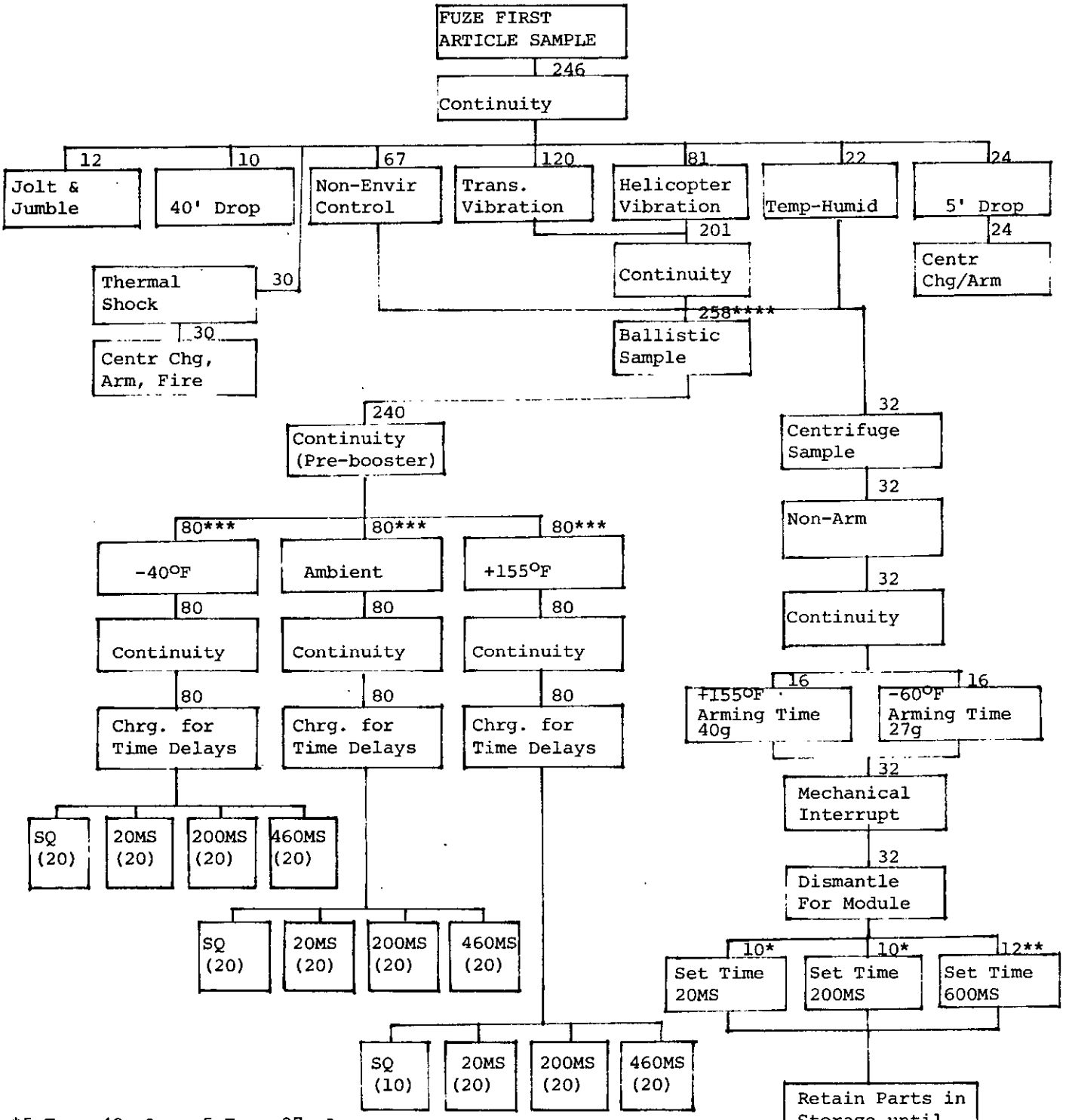
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.41	Environmental Protector		9329847
			NEXT HIGHER ASSEMBLY 9260886

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Holes or cracks in surface		0.40%	3.3	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.14	Visual

NOTES:

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*5 From 40g Arm, 5 From 27g Arm
**6 From 40g Arm, 6 From 27g Arm
***4 From T&H, 16 From Hel-Vibr, 20 From Non-Envir, 40 From Trans. Vibr
****Includes 18 spares from Hel-Vibr to replace launches scored as "No Tests"

NOTE: A setter that represents the helicopter setter system shall be used for delay time acceptance tests in all cases.

TABLE VI First article test

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TABLE VI (Cont'd)

First article sample

Module

1. All modules shall be serialized.

2. A setter that represents the helicopter setter system shall be used for acceptance of all modules.

3. Fixed voltage setter to be used for information on all modules.

4. The same units shall be tested in the same sequence using the same module for both the compensated and fixed voltage sections and data be readily identifiable back to the serialized units. There shall be a one day interval between the compensated and fixed voltage tests with the compensated test preceding the fixed voltage test.

Nominal Time Settings	Environment	Module Quantity	Acceptance Limits
20MS	-60°F	70	5 to 35MSec
	Ambient	70	
	+155°F	70	
	Thermal Shock	24	
200MS	-60°F	70	185 to 215MSec
	Ambient	70	
	+155°F	70	
	Thermal Shock	24	
600MA	-60°F	70	570 to 630MSec
	Ambient	70	
	+155°F	70	The firing capacitor shall be checked to insure that at least minimum detonator firing output pulse is present when the detonator firing circuit is tripped at the conclusion of the time delay.
	Thermal Shock	24	

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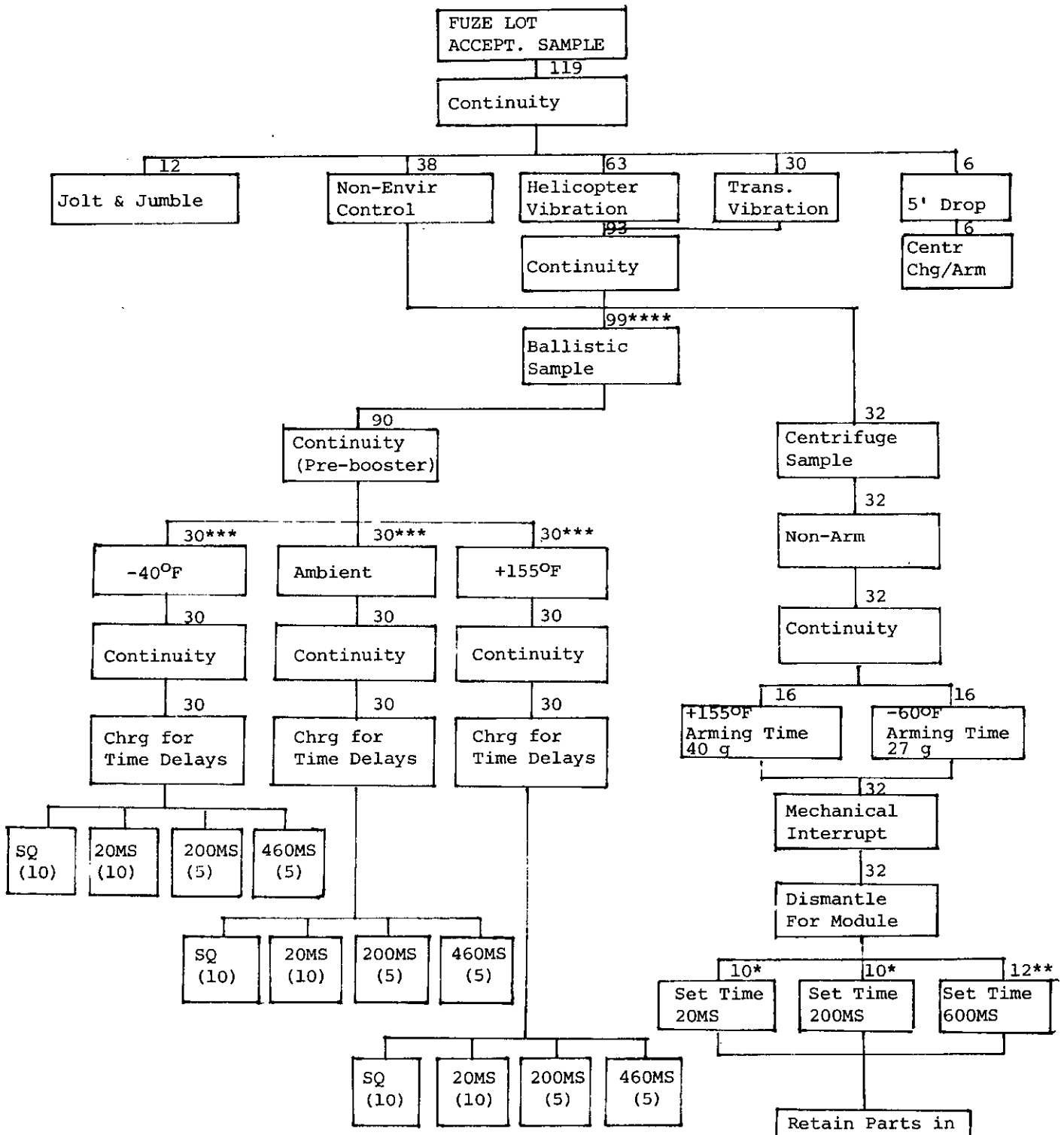
5. Modules shall be temperature conditioned for a minimum of two (2) hours after stabilization at temperature specified in para. 4 above.

6. All modules shall be checked for module turn on time. Data obtained from these checks shall be recorded and be made traceable to individual modules.

7. A seven (7) second time delay shall follow the charging of each module and before the module timing circuit is tripped.

8. Additional nominal time settings (at ambient temperature) shall be used for informational purposes only, the nominal time settings are as follows: 50MSec, 100MSec, 300MSec, 400MSec, 500MSec. Both setters shall be used. Modules that were tested at ambient temperature in the preceding segment (para. 4 above) may be used for this test segment.

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*5 From 40g Arm, 5 From 27g Arm
**6 From 40g Arm, 6 From 27g Arm
***10 From Hel-Vibr, 10 From Non-Envir, 10 From Trans. Vibr.
****Includes 9 spares from Hel-Vibr to replace launches scored as "No Tests"
NOTE: A setter that represents the helicopter setter system shall be used for delay time acceptance in all cases.

TABLE VII Lot acceptance test

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Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1340-A700)

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